An Interpretable Multi-Modal Spam Detection Framework Using Explainable AI and Machine Learning

Sonam Juneja
Computer Science and Engineering
Chandigarh University
Mohali, Punjab, India
sonam.e11002@cumail.in

Vanshit Kumar
Computer Science and Engineering
Chandigarh University
Mohali, Punjab, India
23bcs10087@cuchd.in

Aariz Amaan
Computer Science and Engineering
Chandigarh University
Mohali, Punjab, India
23bcs10422@cuchd.in

Naman
Computer Science and Engineering
Chandigarh University
Mohali, Punjab, India
23bcs11036@cuchd.in

Aditi Arya
Computer Science and Engineering
Chandigarh University
Mohali, Punjab, India
23bcs11535@cuchd.in

Avish Kumar Dubey
Computer Science and Engineering
Chandigarh University
Mohali, Punjab, India
23bcs13898@cuchd.in

***Abstract*—Scam messages reach users through several channels, yet the machine-learning filters that intercept them usually return a verdict without a reason, leaving errors and dataset artefacts invisible. This paper presents an interpretable multi-modal framework that detects fraudulent SMS and e-mail messages and explains every decision. Each channel is served by a specialist pipeline that maps URLs, phone numbers, currency amounts and numbers to readable placeholder tokens, combines uni-/bi-gram TF–IDF features with fifteen stylometric statistics, and fuses Logistic Regression, Random Forest and XGBoost by soft voting weighted with cross-validated F1. LIME explains the full ensemble locally, while exact TreeSHAP attributes the boosted-tree component locally and globally. On held-out splits of the UCI SMS Spam Collection (5,158 messages) and the Enron-Spam corpus (12,000 messages), the ensemble reaches 98.55% accuracy (ROC-AUC 0.9941) on SMS and 98.25% (ROC-AUC 0.9983) on e-mail. Paired McNemar and DeLong tests show that it significantly outperforms both tree learners on e-mail ($p \leq 0.003$, Holm-corrected) and is statistically indistinguishable from Logistic Regression. Deleting the five words ranked highest by LIME lowers the spam probability 5.6–20 times more than deleting random words, supporting explanation faithfulness. Global attributions show that stylometric features dominate SMS decisions and expose a corpus shortcut: the token `enron` occurs in 42% of legitimate test e-mails and in no spam, and appending it to spam e-mails flips 35.5% of the ensemble's detections (91.7% for XGBoost alone). We further show that channel specialisation is essential—cross-channel transfer is near chance—and quantify the cost of heuristic routing.**

***Index Terms*—Spam detection, smishing, phishing, explainable artificial intelligence, SHAP, LIME, ensemble learning, shortcut learning, stylometry.**

I. Introduction

Fraudulent messaging remains the most frequently reported form of cybercrime. In 2025 the FBI's Internet Crime Complaint Center received more than one million complaints with reported losses of US$20.9 billion; phishing and spoofing was again the most reported category (191,561 complaints), and 22,364 complaints involved artificial intelligence (AI) [1]. The threat is changing in character as well as volume. Large language models (LLMs) can generate persuasive, personalised phishing at negligible cost [2]–[4], and scams have followed banking and one-time-password traffic onto SMS ("smishing") [5], [6].

Machine-learning filters have been the standard defence since the first probabilistic spam filters [7], [8], and modern deep and transformer models report excellent benchmark accuracy [9]–[11]. Most deployed filters, however, are black boxes: a message is blocked and neither the user, the analyst nor an auditor can say why. Aggregate accuracy can conceal systematic errors and shortcuts—spurious cues that happen to separate the classes in a particular dataset [12], [13]. At the same time, data-protection and AI legislation has raised expectations of transparency and accountability for automated processing of personal data [14]–[16], and explainable AI (XAI) has matured into a broad toolkit [17]–[19].

Three gaps motivate this work. First, most detectors target a single channel, although SMS and e-mail scams are two faces of one problem with very different statistics (a median of 12 versus 155 words and 12% versus 48% spam in the corpora used here). Second, explanations are usually shown as illustrations of individual predictions rather than evaluated for faithfulness [20], [21] or used to audit what the model has learned. Third, model comparisons on near-saturated benchmarks are rarely accompanied by uncertainty estimates, so differences in the third decimal place are hard to interpret [22].

We address these gaps with an interpretable, multi-modal spam detection framework. Throughout the paper, *modality* denotes the messaging channel (SMS or e-mail); both are text, so the framework is multi-channel rather than multimodal

in the sense of combining text with images or audio. Our contributions are:

1) An interpretability-first pipeline with one specialist per channel: high-signal entities become readable placeholder tokens, TF–IDF features are combined with fifteen named stylometric statistics, and Logistic Regression (LR), Random Forest (RF) and XGBoost (XGB) are fused by performance-weighted soft voting. Training takes about 14 s (SMS) and 77 s (e-mail) on a commodity CPU.
2) A statistically grounded evaluation: bootstrap confidence intervals, exact McNemar and DeLong tests with Holm correction, Matthews correlation coefficient (MCC) and PR-AUC, together with an end-to-end evaluation of the channel router and of cross-channel transfer.
3) A dual explanation layer—LIME over the full ensemble and exact TreeSHAP over the boosted component—whose local faithfulness is tested with a word-deletion experiment.
4) An explanation-driven audit that discovers a corpus shortcut (`enron`), quantifies how far it can be exploited by a one-word insertion, and identifies a subtle sparse-versus-dense pitfall that silently invalidates SHAP explanations of XGBoost text models.

Section II reviews related work, Section III describes the framework, Section IV the experimental protocol, Section V the results, Section VI discusses implications and limitations, and Section VII concludes.

II. Related Work

A. SMS Spam and Smishing Detection

Reproducible SMS research began with the UCI SMS Spam Collection [23]. Recent detectors are predominantly deep or transformer based: Ghourabi and Alohaly combined pretrained transformer embeddings with ensemble learning [24]; Mehmood *et al.* designed a deep smishing detector aimed at reducing false positives [5]; Al-Kabbi *et al.* proposed a two-level feature-fusion filter against fraudulent writing styles [25]; and Nguyen-Xuan and Nguyen introduced a compound-aware encoder-only transformer [26]. Schwarz *et al.* studied smishing from the messaging-platform side, classifying application-to-person traffic by industry [6]. Hur *et al.* examined whether LLMs improve on machine-learning, deep-learning and encoder models for SMS spam [27], while Şahin *et al.* benchmarked classical learners across text representations under imbalanced short-text conditions, emphasising precision and specificity because blocking legitimate messages is costly [28].

Coverage of other languages is growing, including Swahili [29], Dravidian languages with hand-crafted metadata features [30], and cross-lingual detection with GAN-based augmentation for imbalanced data [31]. Robustness is an open concern: Salman *et al.* released a new SMS dataset and analysed spammers' evasive techniques against several models [32], and later evaluated commercial apps, third-party services and open-weight models under perceptible and imperceptible attacks [33]. Rassam and Shaddad studied the robustness and interpretability of naive Bayes and a compact CNN under adversarial attack [34], and SpamDam pursues privacy-preserving, adversary-resistant SMS filtering [35].

B. E-mail Spam and Phishing Detection

E-mail filtering moved from naive Bayes [7], [36] and support vector machines [8] to deep architectures such as hierarchical attention networks [9], hybrid networks with adaptive optimisation [37], vector-similarity search over transformer embeddings [38] and graph attention over content and metadata [39]; a systematic review covers deep phishing-e-mail detectors [40]. Transformers and LLMs now dominate recent work: fine-tuned models for three-way phishing/spam/ham classification [10], comparative fine-tuning of LLMs, NLP models and CNNs [11], few-shot benchmarks against naive Bayes and LightGBM baselines [41], DistilBERT embeddings with neural classifiers [42], comparisons of ML and LLMs with embedded URL analysis [43], and multilingual LLMs for low-resource Kazakh spam [44]. Meléndez *et al.* contrasted traditional models with transformers for phishing e-mail [45]. Ensembles remain competitive through stacking [46], decision fusion [47] and interpretable meta-learners [48]. Few studies cover both channels: Maqsood *et al.* proposed a deep framework for SMS and e-mail [49], and Hadi *et al.* fused shallow lexical features (bag-of-words, TF–IDF, lexical statistics) with deep contextual features [50].

The adversary is also changing. LLMs can author effective phishing at scale [2]; Opara *et al.* evaluated how spam filters handle AI-generated phishing and proposed stylometric detection [51]; detectors must generalise across generator models [52], [53]; persuasion principles in phishing have been quantified with LLMs [54]; and spear phishing has been surveyed comprehensively [55].

C. Explainable AI for Message Filtering

Surveys chart the XAI landscape [17], [18], [56]–[58] and its use in cybersecurity [19], [59]. LIME fits a sparse local surrogate around one prediction [60]; SHAP assigns additive Shapley-value attributions [61] and admits an exact polynomial-time algorithm for tree ensembles [62]. Both are among the most widely used post-hoc tools [58], [63]. In message filtering, explanations have accompanied transformer SMS detectors [64], smishing detectors [65] and transformer phishing models [66]; SHAP has guided feature selection for spam detection [67]; SHAP and LIME have been compared on boosting models for phishing URLs [68]; ChatSpamDetector uses an LLM to produce human-readable rationales [69]; and fuzzy rule bases offer intrinsic interpretability [70].

Explanations are not trustworthy by default. Faithfulness—whether an explanation reflects the model's actual reasoning—must be measured [20], [21], [71]; attention weights are an unreliable proxy [21], [72], [73]; explainers frequently disagree [74], [75]; explanations can be manipulated [76]; and human-centred studies [77] warn that users may over-trust AI advice [78].

D. Shortcut Learning and Dataset Bias

Models may exploit spurious cues that correlate with labels in the training data rather than the intended concept, the "Clever Hans" effect [12], [79]. In NLP such shortcuts originate in dataset biases [80] and persist in LLMs [13]. Spam corpora are vulnerable because legitimate messages are often drawn from a single organisation, so organisation names and internal jargon can become label proxies. Explanations are a natural instrument for exposing such shortcuts [80].

E. Positioning

Table I contrasts representative recent studies. Most address a single channel, and explanations, where present, mainly illustrate individual predictions. We combine two channel specialists with a dual explanation layer, test the faithfulness of the local explanations, use the global explanations to audit the model, and compare learners with paired significance tests.

III. Proposed Framework

Fig. 1 shows the architecture. A message x is routed to the specialist of its channel $c \in \{\text{SMS}, \text{EMAIL}\}$; each specialist has its own normalisation, feature space and trained ensemble, and returns a label $\hat{y} \in \{\text{ham}, \text{spam}\}$, a spam probability and explanations.

A. Interpretability-Preserving Normalisation

Text is lower-cased, HTML tags are stripped, and high-signal entities are *mapped* rather than deleted: URLs become `httpaddr`, e-mail addresses `emailaddr`, currency amounts `moneysymb`, phone numbers `phonenumbr` and remaining numbers `numbr`; other non-letters are removed. Deleting these entities would discard some of the strongest scam signals, whereas readable placeholders can appear verbatim in an explanation. No stemming or lemmatisation is applied, so every token in an explanation is a surface word form a non-expert can recognise.

B. Feature Representation

The normalised text is vectorised with sublinear TF–IDF over uni- and bi-grams [81]. For term t in document d of a training corpus with N documents,

$$w_{t,d} = \left(1 + \ln \mathrm{tf}_{t,d}\right) \left(\ln \frac{1+N}{1+\mathrm{df}_t} + 1\right), \tag{1}$$

after which each document vector is ℓ_2-normalised. Vocabularies are capped at 3,000 terms (SMS, $\mathrm{df} \geq 2$) and 5,000 terms (e-mail, $\mathrm{df} \geq 3$). Fifteen stylometric statistics are computed on the *raw* message (Table II); they are scaled by their training-set maximum absolute value and appended, giving 3,015 (SMS) and 5,015 (e-mail) dimensions. Their names are written in upper case so that they are visually distinct from vocabulary in every explanation.

C. Heterogeneous Base Learners

Three learners with different inductive biases provide ensemble diversity [82]: a linear model, a bagged tree ensemble and a boosted tree ensemble. LR (liblinear, $C = 1$) and RF [83] (250 trees, minimum leaf size 2) use balanced class weights; XGBoost [84] (300 trees, depth 6, learning rate 0.1, row/column subsampling 0.9/0.8, histogram method) uses $\text{scale_pos_weight} = N_{\text{ham}}/N_{\text{spam}}$ on the training split. Class re-weighting was preferred to synthetic oversampling so that every training example remains a real, explainable message. All models use scikit-learn [85] or the XGBoost library with a fixed seed (42).

D. Performance-Weighted Soft Voting

Let $p_m(\text{spam} \mid x)$ be the probability returned by learner m and $\mathrm{F1}_m^{\mathrm{CV}}$ its mean F1 in stratified 5-fold cross-validation on the training split. The ensemble computes

$$p(\text{spam} \mid x) = \sum_{m=1}^{3} w_m \, p_m(\text{spam} \mid x), \tag{2}$$

$$w_m = \frac{\mathrm{F1}_m^{\mathrm{CV}}}{\sum_{k=1}^{3} \mathrm{F1}_k^{\mathrm{CV}}}, \tag{3}$$

and predicts spam when $p(\text{spam} \mid x) \geq 0.5$. The resulting weights are 0.3329/0.3331/0.3340 (LR/RF/XGB) for SMS and 0.3356/0.3311/0.3333 for e-mail: the learners are comparably strong, so any benefit of the ensemble stems from their diversity rather than from a dominant member.

E. Channel Router

At inference time a transparent rule selects the specialist: a message is treated as e-mail if it is longer than 280 characters, contains a line break, or contains one of the cue words *subject*, *dear*, *regards*, *sincerely*, *unsubscribe* or *attached*; otherwise it is treated as SMS. The user can override the choice in the application. Section V-C measures the cost of this heuristic.

F. Explanation Layer

Local, model-agnostic (LIME). LIME [60] perturbs the raw message by removing words, re-scores 1,000 perturbations through the *entire* pipeline (normalisation, features and ensemble) and fits a locally weighted linear surrogate

$$\xi(x) = \arg\min_{g \in \mathcal{G}} \; \mathcal{L}(f, g, \pi_x) + \Omega(g), \tag{4}$$

where f is the ensemble, π_x a proximity kernel and Ω limits the explanation to the ten most influential words. Because LIME queries the deployed pipeline, it explains the complete ensemble.

Exact attributions (TreeSHAP). For the XGBoost component, SHAP values ϕ_j decompose the log-odds output additively, $f_{\text{XGB}}(x) = \phi_0 + \sum_j \phi_j(x)$ [61], and are computed exactly in polynomial time by TreeSHAP [62]. Global importance is the mean absolute attribution over a set S of 800 held-out messages, $I_j = \frac{1}{|S|} \sum_{x \in S} |\phi_j(x)|$.

Implementation pitfall. XGBoost treats entries that are *not stored* in a sparse matrix as missing values, whereas a dense

TABLE I
REPRESENTATIVE RECENT STUDIES COMPARED WITH THIS WORK

Study	Year	Channel	Approach	Explanation
Ghourabi & Alohaly [24]	2023	SMS	Transformer embeddings + ensemble	–
Maqsood *et al.* [49]	2023	SMS + e-mail	Deep learning framework	–
Salman *et al.* [32]	2024	SMS	ML/DL models under evasive techniques; new dataset	–
Roumeliotis *et al.* [11]	2024	E-mail	Fine-tuned LLMs, NLP models and CNNs	–
Meléndez *et al.* [45]	2024	E-mail (phishing)	Traditional ML vs. transformers	–
Uddin *et al.* [64]	2025	SMS	Transformer language model	Post-hoc explainability analysis
Kshirsagar *et al.* [48]	2025	E-mail	Meta-learner ensembles	Interpretable by design
Uddin *et al.* [66]	2026	E-mail (phishing)	Fine-tuned transformer	Post-hoc explanations
Koide *et al.* [69]	2026	E-mail (phishing)	LLM-based detection	LLM-generated rationale
Jasim *et al.* [70]	2026	SMS	Fuzzy logic + BiLSTM	Fuzzy rules
Prince *et al.* [65]	2026	SMS (smishing)	Hybrid transformer feature fusion	Post-hoc XAI
Hadi *et al.* [50]	2026	SMS + e-mail	Shallow lexical + deep contextual features	–
This work	2026	SMS + e-mail (routed specialists)	TF–IDF + stylometry; LR, RF, XGB fused by F1-weighted soft voting	LIME (full ensemble) + exact TreeSHAP; deletion faithfulness test; shortcut audit

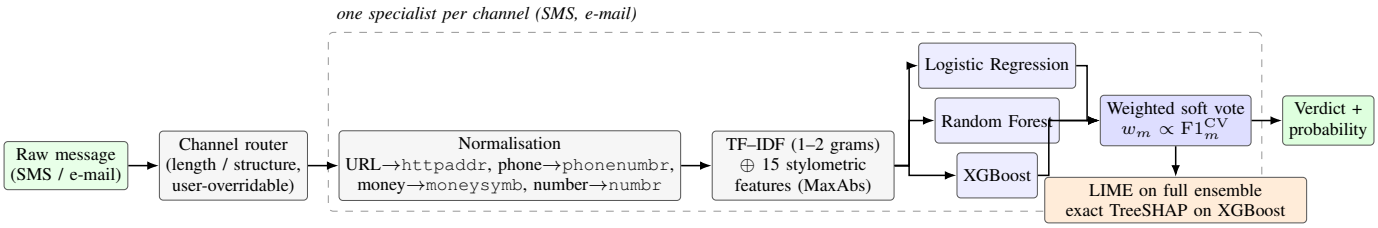


Fig. 1. Architecture of the proposed framework. Each channel has its own normalisation, feature space and ensemble; the explanation layer returns a local explanation with every verdict and global summaries of each model.

TABLE II
ENGINEERED STYLOMETRIC FEATURES (COMPUTED ON THE RAW MESSAGE)

Feature	Description
`CHAR_COUNT`, `WORD_COUNT`	Length in characters / words
`AVG_WORD_LEN`	Mean word length
`DIGIT_RATIO`	Share of characters that are digits
`UPPERCASE_RATIO`	Share of letters in upper case
`ALLCAPS_WORDS`	Words written fully in capitals
`PUNCT_RATIO`	Share of punctuation characters
`SPECIAL_CHAR_RATIO`	Share of non-alphanumeric characters
`NUM_EXCLAMATIONS`	Count of "!"
`NUM_QUESTIONS`	Count of "?"
`NUM_CURRENCY`	Currency symbols and money amounts
`NUM_URLS`, `NUM_EMAILS`	Detected links / e-mail addresses
`NUM_PHONES`	Detected phone numbers
`SPAM_KEYWORDS`	Hits in a 46-word, human-auditable scam lexicon

array turns them into genuine zeros that follow a different branch. Explaining a model trained on sparse TF–IDF matrices with a densified copy of the input therefore explains a different function: in our pipeline, densification changes the XGBoost verdict for 729 of 1,032 SMS and 1,182 of 2,400 e-mail test messages. All attributions in this paper are computed with XGBoost's native TreeSHAP on exactly the sparse matrix the model scores, so that they sum to the deployed model's output.

TABLE III
CORPORA AFTER CLEANING AND DE-DUPLICATION

Channel	Messages	Spam	Median words	Train	Test
SMS [23]	5,158	642 (12.4%)	12	4,126	1,032
E-mail [36]	12,000	5,792 (48.3%)	155	9,600	2,400

IV. EXPERIMENTAL SETUP

A. Datasets

Table III summarises the corpora after removing empty and duplicate messages. The UCI SMS Spam Collection [23] is used in full. For e-mail we use the Enron-Spam corpus [36] (subject and body, truncated to 8,000 characters), drawing a class-stratified sample of 12,000 messages from the de-duplicated corpus to keep the full pipeline laptop-friendly. The channels differ sharply in length and class balance, which motivates channel-specific models.

B. Protocol and Metrics

Each corpus is split 80/20 with stratification (seed 42). Vectorisers, scalers, cross-validation weights and models are

fitted on the training split only, and the test split is used once for final evaluation. We report accuracy, precision, recall, F1, MCC (preferred for imbalanced binary problems [86]), ROC-AUC, PR-AUC and the Brier score of the predicted probabilities.

C. Statistical Analysis

Uncertainty is quantified with 95% percentile bootstrap confidence intervals over 2,000 resamples of the test set [87]. The ensemble is compared with each base learner on the *same* test messages using the exact McNemar test on discordant predictions [22], [88] and DeLong's test for correlated ROC curves [89]; p-values are Holm-corrected within each channel [90].

D. Additional Analyses

Routing and transfer. The router is evaluated on the pooled test sets (3,432 messages) against oracle routing, and each specialist is applied to the other channel's test set. *Explanation faithfulness.* Following the comprehensiveness idea of ERASER [71], for correctly detected spam messages we remove all occurrences of the top-k words ranked by LIME ($k \in \{1, 3, 5\}$) and record the drop in ensemble spam probability, compared with removing k random words (10 draws). We use 60 SMS and 25 e-mail messages because each explanation requires 1,000 pipeline evaluations. *Shortcut probe.* For the token surfaced by the global attributions we measure (i) deletion of the token from legitimate test e-mails and (ii) insertion of a single occurrence into spam test e-mails.

E. Implementation

The framework is implemented in Python 3.12 with scikit-learn, XGBoost, SHAP and LIME, and served through a four-tab Streamlit application (detector, model performance, global explanations, documentation). A 13-check automated smoke test covers data loading, both detectors, LIME, SHAP and a simulated application session. All timings were measured on a 10-core ARM64 CPU without a GPU; re-running the training code reproduces the reported metrics exactly.

V. Results

A. Classification Performance

Table IV reports all four systems on both channels and Fig. 2 shows the ROC curves. The ensemble reaches 98.55% accuracy on SMS and 98.25% on e-mail. On SMS it attains the highest precision (0.9669), the lowest false-positive rate (4 of 904 legitimate messages, tied with XGBoost), the best PR-AUC (0.9790) and the best Brier score (0.0130), and its ROC-AUC (0.9941) ties the best base learners to four decimals. On e-mail it has the highest recall (0.9940; 7 of 1,158 spam messages missed) and the highest ROC-AUC (0.9983) and PR-AUC (0.9980). LR, however, is a remarkably strong baseline: it matches the ensemble's SMS accuracy and slightly exceeds it in SMS F1 and in e-mail accuracy and F1. The confidence intervals of all systems overlap substantially on SMS, where the test set contains only 128 spam messages.

The two specialists err in opposite directions: the SMS ensemble favours precision (4 false positives, 11 false negatives), the e-mail ensemble favours recall (35 false positives, 7 false negatives). On SMS, false alarms are costly because a suppressed message may be a one-time password, so this profile suits the channel.

B. Statistical Significance

Table V lists the paired tests. On SMS no difference is significant: the ensemble and each base learner disagree on at most eight messages. On e-mail the ensemble is significantly better than both tree models on discordant predictions (McNemar: 24 vs. 5 messages against RF and 22 vs. 4 against XGB; Holm-corrected $p = 0.002$) and in ROC-AUC (DeLong, Holm-corrected $p \leq 0.003$), but it is statistically indistinguishable from LR (11 vs. 13 discordant messages, $p = 0.84$). Combining a linear model with tree ensembles therefore lifts the tree models to LR's level and improves the ranking metrics, but on these corpora it does not surpass a well-regularised linear model on TF–IDF features.

C. Routing and Cross-Channel Transfer

The specialists do not transfer (Table VI). Applied to e-mail, the SMS model is at chance level (ROC-AUC 0.53, MCC 0.04); applied to SMS, the e-mail model labels 88% of legitimate messages as spam. Channel-specific training is therefore essential. The router assigns 96.4% of SMS and 85.1% of e-mail test messages to the correct specialist; misrouted e-mails are short messages without cue words, and the SMS specialist classifies only 42.3% of them correctly. Consequently, end-to-end accuracy on the pooled test sets drops from 98.34% with oracle routing to 92.16% with automatic routing. The classifiers are accurate; the heuristic router is the weakest component of the deployed system, which is why the application exposes a manual override.

D. Local Explanations

Fig. 3 explains an illustrative smishing message composed by the authors and absent from both corpora: "URGENT! You have WON a £2,000 cash prize GUARANTEED. Call 09061701461 now or visit www.claim-your-prize.com to claim!" All three learners vote spam (LR 1.000, RF 0.987, XGB 1.000; ensemble 0.996). LIME ranks the premium-rate phone number highest, followed by the money amount and scam vocabulary (*claim*, *URGENT*, *prize*). The exact Tree-SHAP decomposition of the XGBoost component attributes the decision mainly to stylometry—the digit ratio, mean word length, scam-lexicon hits and capitalisation—together with the `httpaddr` and `phonenumbr` placeholders. The two views are complementary: LIME speaks in the user's words, SHAP in the model's features.

E. Global Explanations: Stylometry Rivals Vocabulary

Fig. 4 ranks features by mean absolute attribution over 800 held-out messages per channel. On SMS, eight of the ten most important features are stylometric, led by

TABLE IV
HELD-OUT TEST PERFORMANCE WITH 95% BOOTSTRAP CONFIDENCE INTERVALS (SMS: $n = 1{,}032$; E-MAIL: $n = 2{,}400$). LOWER BRIER IS BETTER.

Channel	Model	Accuracy	Precision	Recall	F1	MCC	ROC-AUC	PR-AUC	Brier
SMS	LR	0.9855 [.978, .992]	0.9520 [.911, .984]	0.9297 [.882, .970]	0.9407 [.910, .967]	0.9325 [.898, .963]	0.9941 [.989, .998]	0.9768 [.959, .992]	0.0148 [.011, .019]
	RF	0.9845 [.977, .991]	0.9516 [.905, .985]	0.9219 [.869, .965]	0.9365 [.902, .964]	0.9278 [.890, .959]	0.9929 [.985, .998]	0.9770 [.958, .992]	0.0184 [.015, .023]
	XGB	0.9845 [.977, .991]	0.9667 [.928, .992]	0.9062 [.849, .954]	0.9355 [.902, .963]	0.9273 [.890, .959]	0.9941 [.989, .998]	0.9739 [.954, .989]	0.0135 [.008, .021]
	Ensemble	**0.9855** [.978, .992]	**0.9669** [.929, .992]	**0.9141** [.860, .960]	**0.9398** [.907, .967]	**0.9320** [.895, .962]	**0.9941** [.988, .999]	**0.9790** [.961, .993]	**0.0130** [.009, .018]
E-mail	LR	0.9833 [.978, .988]	0.9753 [.966, .984]	0.9905 [.984, .996]	0.9829 [.978, .988]	0.9668 [.957, .977]	0.9979 [.997, .999]	0.9976 [.996, .999]	0.0210 [.018, .024]
	RF	0.9746 [.968, .980]	0.9629 [.951, .973]	0.9853 [.978, .992]	0.9740 [.967, .980]	0.9494 [.937, .961]	0.9973 [.996, .998]	0.9970 [.996, .998]	0.0380 [.035, .041]
	XGB	0.9750 [.969, .981]	0.9645 [.954, .975]	0.9845 [.977, .991]	0.9744 [.968, .981]	0.9502 [.938, .963]	0.9969 [.996, .998]	0.9965 [.995, .998]	0.0189 [.015, .023]
	Ensemble	**0.9825** [.977, .988]	**0.9705** [.961, .980]	**0.9940** [.989, .998]	**0.9821** [.977, .987]	**0.9652** [.955, .975]	**0.9983** [.997, .999]	**0.9980** [.997, .999]	**0.0213** [.019, .024]

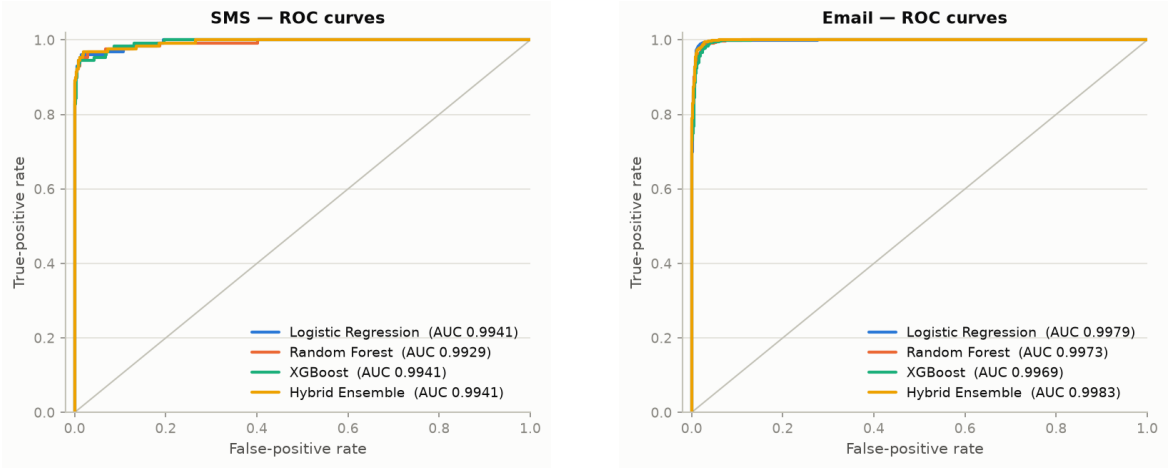


Fig. 2. ROC curves of the four systems on the held-out SMS (left) and e-mail (right) test sets. All systems exceed 0.99 AUC; differences are resolved by the paired tests in Table V.

TABLE V
PAIRED TESTS: ENSEMBLE VS. BASE LEARNERS. b/c: MESSAGES ONLY THE ENSEMBLE / ONLY THE BASE LEARNER CLASSIFIES CORRECTLY. p-VALUES HOLM-CORRECTED.

		McNemar		DeLong (ROC-AUC)		
Ch.	**Comparison**	b/c	p	ΔAUC	z	p
SMS	Ens. vs. LR	4/4	1.000	+0.0000	0.02	1.000
	Ens. vs. RF	2/1	1.000	+0.0012	1.15	0.754
	Ens. vs. XGB	1/0	1.000	-0.0000	-0.01	1.000
E-mail	Ens. vs. LR	11/13	0.839	+0.0004	1.26	0.208
	Ens. vs. RF	24/5	0.002	+0.0009	3.14	0.003
	Ens. vs. XGB	22/4	0.002	+0.0014	3.66	$<$0.001

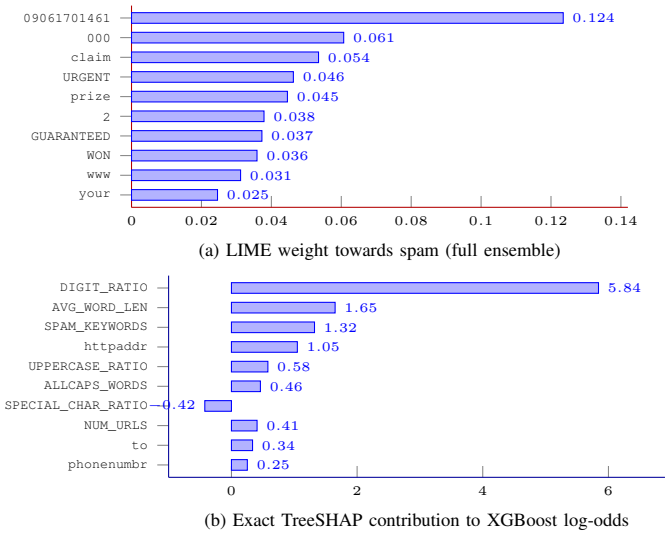
TABLE VI
ROUTING AND CROSS-CHANNEL TRANSFER OF THE ENSEMBLES (POOLED TEST SET: 3,432 MESSAGES)

Setting	**Acc.**	**F1**	**MCC**	**AUC**	**FPR**
Oracle routing (pooled)	0.9834	0.9780	0.9648	0.9978	0.0182
Automatic routing (pooled)	0.9216	0.8921	0.8317	0.9687	0.0443
SMS model $\rightarrow$ e-mail test	0.5208	0.5287	0.0442	0.5290	0.5129
E-mail model $\rightarrow$ SMS test	0.2209	0.2299	0.0594	0.6707	0.8805

DIGIT_RATIO, AVG_WORD_LEN and SPAM_KEYWORDS; the fifteen engineered features (0.5% of the dimensions) carry 74.5% of the total attribution mass. A short scam is betrayed by its *shape*—digits, long words, capitals and punctuation—as much as by its vocabulary. On e-mail the ranking is dominated by a single word, enron, whose mean attribution is about three times that of the next feature; stylometric features (NUM_EXCLAMATIONS, CHAR_COUNT, DIGIT_RATIO) and correspondence vocabulary (*thanks*, *attached*) follow.

F. Faithfulness of the Local Explanations

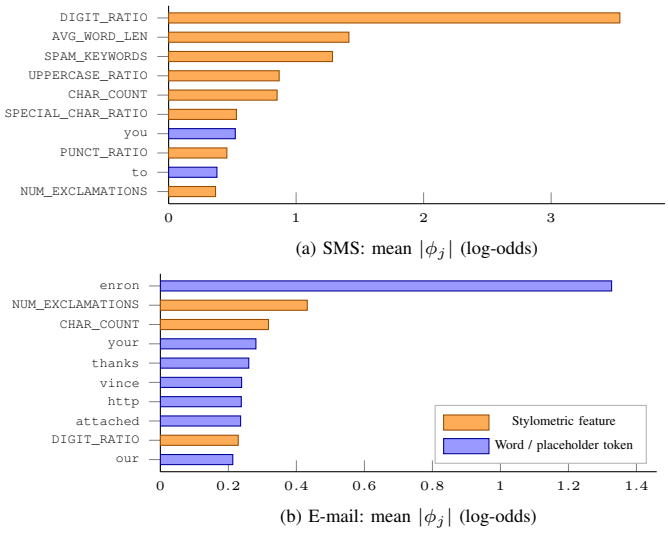
Table VII reports the deletion test. Removing the single highest-ranked LIME word lowers the ensemble's spam probability by 0.106 on average for SMS, 8.6 times more than removing a random word, and removing the top five words lowers it by 0.455 and flips 52% of the verdicts. For e-mail the drops are smaller in absolute terms—long messages



(a) LIME weight towards spam (full ensemble)

(b) Exact TreeSHAP contribution to XGBoost log-odds

Fig. 3. Local explanations of the illustrative smishing SMS (ensemble spam probability 0.996): (a) LIME weights over the full ensemble; (b) the ten largest exact TreeSHAP contributions of the XGBoost component (base value 0.50 log-odds). Upper-case names are stylometric features.



(a) SMS: mean $|\phi_j|$ (log-odds)

(b) E-mail: mean $|\phi_j|$ (log-odds)

Fig. 4. Global importance (exact TreeSHAP, XGBoost component, 800 held-out messages per channel): the ten most important features for (a) SMS and (b) e-mail.

TABLE VII
DELETION TEST OF LIME FAITHFULNESS: MEAN DROP IN ENSEMBLE SPAM PROBABILITY AFTER REMOVING THE TOP-k LIME WORDS / k RANDOM WORDS

Channel	$k=1$	$k=3$	$k=5$	**Flips** (k=5)
SMS ($n=60$)	0.106 / 0.012	0.291 / 0.043	0.455 / 0.081	52%
E-mail ($n=25$)	0.043 / 0.002	0.133 / 0.005	0.198 / 0.010	24%

contain redundant evidence—but about 20 times larger than for random deletions. LIME's rankings therefore identify words on which the ensemble's decision genuinely depends.

TABLE VIII
SHORTCUT PROBE ON THE E-MAIL TEST SET: INSERTING "ENRON" INTO DETECTED SPAM AND DELETING IT FROM LEGITIMATE MAIL

	Insertion into spam		**Deletion from ham**	
Model	**Flips to ham**	$\overline{\Delta p}$	**Flips**	$\overline{\Delta p}$
LR	52/1147 (4.5%)	-0.066	2/523	+0.032
RF	289/1141 (25.3%)	-0.301	10/523	+0.053
XGB	1045/1140 (91.7%)	-0.763	19/523	+0.052
Ensemble	409/1151 (35.5%)	-0.376	11/523	+0.046

G. Explanations as an Audit Tool: A Corpus Shortcut

The prominence of `enron` prompted an audit. The token occurs in 44.4% of legitimate training e-mails and 42.1% of legitimate test e-mails, but in *none* of the spam messages in either split: every legitimate e-mail in the corpus is internal Enron correspondence, so the company name is a perfect one-sided proxy for the ham class. The attributions confirm that the model relies on it: its mean TreeSHAP contribution is -3.95 log-odds (towards ham) when the token is present, versus $+0.66$ when it is absent. Other identity cues of the same organisation—for example the first name *vince*—also rank among the most important features. Notably, *none* of the ensemble's 35 false positives contains `enron`: the legitimate e-mails that are misclassified are precisely those lacking the corpus marker.

Table VIII quantifies the consequences. Deleting the token from 523 legitimate test e-mails raises the ensemble's spam probability only slightly (11 verdicts flip), because legitimate correspondence contains other ham evidence. The insertion direction, however, exposes a practical evasion: appending the single word "enron" to spam e-mails flips 409 of the 1,151 detected spam messages (35.5%) to ham. Vulnerability differs sharply between learners—XGBoost is evaded on 91.7% of messages, RF on 25.3% and LR on only 4.5%—so the heterogeneous ensemble is substantially more robust than its boosted member, yet far from immune. This shortcut is invisible in Tables IV–V, where the e-mail model appears near-perfect; it became visible only through the global explanations.

H. Computational Cost

Table IX summarises the cost. Training both specialists, including 5-fold cross-validation of all three learners, takes about 1.5 minutes on a CPU. A single verdict takes about 15 ms, an exact TreeSHAP attribution under a millisecond, and a LIME explanation with 1,000 perturbations 0.07–0.21 s, so every prediction can be explained interactively. Batched scoring processes thousands of messages per second. Such task-specific models avoid the energy cost associated with large general-purpose generative models [91].

VI. DISCUSSION

Accuracy is not the bottleneck. On both corpora all learners exceed 97% accuracy and 0.99 ROC-AUC, and the paired tests show that a regularised linear model on well-designed features is as accurate as the ensemble. This agrees with recent

TABLE IX
COMPUTATIONAL COST ON A 10-CORE ARM64 CPU (NO GPU)

Channel	Train (s)	Verdict (ms)	Batch (msg/s)	SHAP (ms)	LIME (ms)
SMS	14.3	14.7	14,897	0.86	76
E-mail	76.8	14.8	1,967	0.75	210

findings that feature representation matters as much as model complexity for short-text spam [28] and that classical models remain competitive with transformers on phishing e-mail [45]. The ensemble's value lies in improved ranking metrics, a better precision profile on SMS, significantly better performance than either tree model on e-mail and—most clearly—robustness to the shortcut attack, which the boosted model alone does not have.

Explanations change what the engineer can see. The `enron` shortcut is a property of the corpus, not a defect of the pipeline, but a black-box deployment would have shipped it silently, and an attacker who learned of it could evade one third of detections with a single word. This echoes concerns about shortcut learning in NLP [13], [80] and supports treating global explanations as an auditing instrument. Explanations must themselves be validated: the sparse/dense pitfall described in Section III produces plausible-looking SHAP plots of a function the deployed model does not compute, illustrating why faithfulness has to be tested [20], [21] rather than assumed.

Practical guidance. We recommend (i) computing tree attributions on the exact input representation the model scores; (ii) validating local explanations with deletion tests; (iii) inspecting global attributions for organisation names, dates and other corpus identifiers before deployment; and (iv) evaluating the routing step, not only the classifiers, in multi-channel systems.

Limitations. Both corpora are English and date from 2000–2012; modern smishing, multilingual and code-mixed messages, and LLM-generated phishing [2], [51] are not represented, and the Enron-Spam shortcut means e-mail accuracy overstates performance on other organisations' mail. Results come from a single stratified split (complemented by cross-validation on the training data), the router is a hand-written heuristic, SHAP explains only the boosted component, and the faithfulness test uses a moderate number of messages. We did not benchmark a fine-tuned transformer or an LLM under the same protocol, and adversarial robustness beyond the shortcut probe [32], [33] was not evaluated.

VII. CONCLUSION AND FUTURE WORK

We presented an interpretable multi-modal framework that detects scam SMS and e-mail messages with channel-specific ensembles of Logistic Regression, Random Forest and XGBoost and explains every verdict with LIME and exact TreeSHAP. The ensemble reaches 98.55% accuracy on SMS and 98.25% on e-mail, significantly outperforms both tree learners on e-mail, and trains in minutes on a CPU. A deletion test confirms that its local explanations are faithful, and its global explanations show that stylometry rivals vocabulary on SMS and expose a corpus shortcut that a one-word insertion can exploit—an insight no accuracy table revealed. Future work will (i) evaluate cross-corpus generalisation and remove identity shortcuts by masking or counterfactual augmentation, (ii) extend the framework to multilingual and code-mixed corpora, including Indian languages, (iii) replace the heuristic router with a learned channel classifier, (iv) add probability calibration and counterfactual explanations, (v) benchmark transformer and LLM detectors under the same statistical and explanation protocol, and (vi) test robustness against obfuscation and LLM-generated scams.

REFERENCES

[1] Federal Bureau of Investigation, "Internet crime report 2025," Internet Crime Complaint Center (IC3), Washington, DC, USA, Tech. Rep., Apr. 2026, [Online]. Available: https://www.ic3.gov/AnnualReport/Reports/2025_IC3Report.pdf.

[2] F. Heiding, B. Schneier, A. Vishwanath, J. Bernstein, and P. S. Park, "Devising and detecting phishing emails using large language models," *IEEE Access*, vol. 12, pp. 42 131–42 146, 2024, doi: 10.1109/ACCESS.2024.3375882.

[3] Y. Yao, J. Duan, K. Xu, Y. Cai, Z. Sun, and Y. Zhang, "A survey on large language model (llm) security and privacy: The good, the bad, and the ugly," *High-Confidence Computing*, vol. 4, no. 2, Jun. 2024, art. no. 100211, doi: 10.1016/j.hcc.2024.100211.

[4] H. Xu *et al.*, "Large language models for cyber security: A systematic literature review," *ACM Transactions on Software Engineering and Methodology*, Sep. 2025, art. no. 3769676, doi: 10.1145/3769676.

[5] M. K. Mehmood, H. Arshad, M. Alawida, and A. Mehmood, "Enhancing smishing detection: A deep learning approach for improved accuracy and reduced false positives," *IEEE Access*, vol. 12, pp. 137 176–137 193, 2024, doi: 10.1109/ACCESS.2024.3463871.

[6] S. F. Schwarz, P. Fonseca, and A. Rocha, "Smishing detection from a messaging platform view," *IEEE Access*, vol. 13, pp. 143 449–143 464, 2025, doi: 10.1109/ACCESS.2025.3597903.

[7] M. Sahami, S. Dumais, D. Heckerman, and E. Horvitz, "A Bayesian approach to filtering junk e-mail," in *Proc. AAAI Workshop on Learning for Text Categorization*, Madison, WI, USA, 1998, tech. Rep. WS-98-05.

[8] H. Drucker, D. Wu, and V. N. Vapnik, "Support vector machines for spam categorization," *IEEE Transactions on Neural Networks*, vol. 10, no. 5, pp. 1048–1054, Sep. 1999, doi: 10.1109/72.788645.

[9] S. Zavrak and S. Yilmaz, "Email spam detection using hierarchical attention hybrid deep learning method," *Expert Systems with Applications*, vol. 233, Dec. 2023, art. no. 120977, doi: 10.1016/j.eswa.2023.120977.

[10] S. Jamal, H. Wimmer, and I. H. Sarker, "An improved transformer-based model for detecting phishing, spam and ham emails: A large language model approach," *SECURITY AND PRIVACY*, vol. 7, no. 5, Sep. 2024, art. no. e402, doi: 10.1002/spy2.402.

[11] K. I. Roumeliotis, N. D. Tselikas, and D. K. Nasiopoulos, "Next-generation spam filtering: Comparative fine-tuning of llms, nlps, and cnn models for email spam classification," *Electronics*, vol. 13, no. 11, May 2024, art. no. 2034, doi: 10.3390/electronics13112034.

[12] R. Geirhos *et al.*, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020, doi: 10.1038/s42256-020-00257-z.

[13] M. Du, F. He, N. Zou, D. Tao, and X. Hu, "Shortcut learning of large language models in natural language understanding," *Communications of the ACM*, vol. 67, no. 1, pp. 110–120, Jan. 2024, doi: 10.1145/3596490.

[14] European Parliament and Council of the European Union, "Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data (General Data Protection Regulation)," Official Journal of the European Union, L 119, pp. 1–88, May 2016.

[15] ——, "Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)," Official Journal of the European Union, L series, Jul. 2024.

[16] Government of India, "The digital personal data protection act, 2023 (Act no. 22 of 2023)," The Gazette of India, Ministry of Law and Justice, Aug. 2023.

[17] S. Ali *et al.*, "Explainable artificial intelligence (xai): What we know and what is left to attain trustworthy artificial intelligence," *Information Fusion*, vol. 99, Nov. 2023, art. no. 101805, doi: 10.1016/j.inffus.2023.101805.

[18] L. Longo *et al.*, "Explainable artificial intelligence (xai) 2.0: A manifesto of open challenges and interdisciplinary research directions," *Information Fusion*, vol. 106, Jun. 2024, art. no. 102301, doi: 10.1016/j.inffus.2024.102301.

[19] G. Rjoub *et al.*, "A survey on explainable artificial intelligence for cybersecurity," *IEEE Transactions on Network and Service Management*, vol. 20, no. 4, pp. 5115–5140, Dec. 2023, doi: 10.1109/TNSM.2023.3282740.

[20] M. Nauta *et al.*, "From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable ai," *ACM Computing Surveys*, vol. 55, no. 13s, pp. 1–42, Dec. 2023, doi: 10.1145/3583558.

[21] Q. Lyu, M. Apidianaki, and C. Callison-Burch, "Towards faithful model explanation in nlp: A survey," *Computational Linguistics*, vol. 50, no. 2, pp. 657–723, Jun. 2024, doi: 10.1162/coli_a_00511.

[22] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998, doi: 10.1162/089976698300017197.

[23] T. A. Almeida, J. M. G. Hidalgo, and A. Yamakami, "Contributions to the study of SMS spam filtering: New collection and results," in *Proc. 11th ACM Symp. Document Engineering (DocEng)*, Mountain View, CA, USA, 2011, pp. 259–262, doi: 10.1145/2034691.2034742.

[24] A. Ghourabi and M. Alohaly, "Enhancing spam message classification and detection using transformer-based embedding and ensemble learning," *Sensors*, vol. 23, no. 8, Apr. 2023, art. no. 3861, doi: 10.3390/s23083861.

[25] H. A. Al-Kabbi, M.-R. Feizi-Derakhshi, and S. Pashazadeh, "A hierarchical two-level feature fusion approach for sms spam filtering," *Intelligent Automation & Soft Computing*, vol. 39, no. 4, pp. 665–682, 2024, doi: 10.32604/iasc.2024.050452.

[26] S. Nguyen-Xuan and H. Nguyen, "A compound-aware encoder-only transformer model for smishing detection," *The Journal of Supercomputing*, vol. 82, no. 4, Feb. 2026, art. no. 176, doi: 10.1007/s11227-026-08321-y.

[27] M. Hur, S. Seo, J. Hwang, and M. Min, "Are large language models effective for detecting spam messages?" *International Journal of Information Security*, vol. 25, no. 4, Jul. 2026, art. no. 127, doi: 10.1007/s10207-026-01283-5.

[28] M. S. Şahin, D. Özkan Şahin, and A. F. Salah, "Revisiting sms spam detection: The impact of feature representation on classical machine learning models," *Electronics*, vol. 15, no. 4, Feb. 2026, art. no. 894, doi: 10.3390/electronics15040894.

[29] I. S. Mambina, J. D. Ndibwile, D. Uwimpuhwe, and K. F. Michael, "Uncovering sms spam in swahili text using deep learning approaches," *IEEE Access*, vol. 12, pp. 25 164–25 175, 2024, doi: 10.1109/ACCESS.2024.3365193.

[30] E. Ramanujam, A. M. Abirami, K. Sakthiprakash, and S. Sumitra, "Efficient extraction and evaluation of hand-crafted meta-data features for dravidian spam sms classification," *Evolving Systems*, vol. 17, no. 1, Feb. 2026, art. no. 8, doi: 10.1007/s12530-025-09761-2.

[31] A. Filali *et al.*, "Cross-lingual sms spam detection using gan-based augmentation for imbalanced datasets," *Scientific Reports*, vol. 16, no. 1, Feb. 2026, art. no. 7128, doi: 10.1038/s41598-026-37769-4.

[32] M. Salman, M. Ikram, and M. A. Kaafar, "Investigating evasive techniques in sms spam filtering: A comparative analysis of machine learning models," *IEEE Access*, vol. 12, pp. 24 306–24 324, 2024, doi: 10.1109/ACCESS.2024.3364671.

[33] M. Salman, M. Islam, M. Ikram, and M. A. Kaafar, "Johnny still receives spam sms: Assessing the robustness of sms spam detection," 2026, arXiv:2609.01171.

[34] M. A. Rassam and R. Shaddad, "Analysis of robustness and interpretability of multinomial naïve bayes and tiny text cnn models for sms spam detection under adversarial attacks," *Information*, vol. 17, no. 5, Apr. 2026, art. no. 408, doi: 10.3390/info17050408.

[35] Y. Li, R. Zhang, W. Rong, and X. Mi, "Spamdam: Towards privacy-preserving and adversary-resistant sms spam detection," 2024, arXiv:2404.09481.

[36] V. Metsis, I. Androutsopoulos, and G. Paliouras, "Spam filtering with naive Bayes—which naive Bayes?" in *Proc. 3rd Conf. Email and Anti-Spam (CEAS)*, Mountain View, CA, USA, 2006.

[37] M. Hosseinzadeh *et al.*, "Improving phishing email detection performance through deep learning with adaptive optimization," *Scientific Reports*, vol. 15, no. 1, Oct. 2025, art. no. 36724, doi: 10.1038/s41598-025-20668-5.

[38] C. Patra, D. Giri, S. Nandi, A. K. Das, and M. J. F. Alenazi, "Phishing email detection using vector similarity search leveraging transformer-based word embedding," *Computers and Electrical Engineering*, vol. 124, May 2025, art. no. 110403, doi: 10.1016/j.compeleceng.2025.110403.

[39] M. Safran and A. Musleh, "Phishinggnn: Phishing email detection using graph attention networks and transformer-based feature extraction," *IEEE Access*, vol. 13, pp. 131 390–131 399, 2025, doi: 10.1109/ACCESS.2025.3592135.

[40] K. Thakur, M. L. Ali, M. A. Obaidat, and A. Kamruzzaman, "A systematic review on deep-learning-based phishing email detection," *Electronics*, vol. 12, no. 21, Nov. 2023, art. no. 4545, doi: 10.3390/electronics12214545.

[41] M. Labonne and S. Moran, "Spam-t5: Benchmarking large language models for few-shot email spam detection," 2023, arXiv:2304.01238.

[42] G. P. Reddy, R. Dsouza, Ramyashree, S. Raghavendra, and P. S. Venugopala, "An llm driven framework for email spam detection using distilbert embeddings and neural classifiers," *Discover Applied Sciences*, vol. 8, no. 2, Dec. 2025, art. no. 146, doi: 10.1007/s42452-025-08147-y.

[43] S. Tarapiah, L. Abbas, O. Mardawi, S. Atalla, Y. Himeur, and W. Mansoor, "Evaluating the effectiveness of large language models (llms) versus machine learning (ml) in identifying and detecting phishing email attempts," *Algorithms*, vol. 18, no. 10, Sep. 2025, art. no. 599, doi: 10.3390/a18100599.

[44] K. Mukhammed-Ali and S. Assem, "Applying large language models to spam detection in the kazakh low-resource language setting," *Scientific Reports*, vol. 16, no. 1, Jun. 2026, art. no. 30145, doi: 10.1038/s41598-026-60445-6.

[45] R. Meléndez, M. Ptaszynski, and F. Masui, "Comparative investigation of traditional machine-learning models and transformer models for phishing email detection," *Electronics*, vol. 13, no. 24, Dec. 2024, art. no. 4877, doi: 10.3390/electronics13244877.

[46] Y. Tian, X. Dai, Z. Li, H. Guo, and X. Mao, "Improving the accuracy of cybersecurity spam email detection using ensemble techniques: A stacking approach machine learning for spam email detection," *PLOS One*, vol. 20, no. 9, Sep. 2025, art. no. e0331574, doi: 10.1371/journal.pone.0331574.

[47] M. Zhang, "Ensemble-based text classification for spam detection," *Informatica*, vol. 48, no. 6, Apr. 2024, doi: 10.31449/inf.v48i6.5246.

[48] M. Kshirsagar, V. Rathi, and C. Ryan, "Meta-learner-based frameworks for interpretable email spam detection," *Frontiers in Artificial Intelligence*, vol. 8, Oct. 2025, art. no. 1569804, doi: 10.3389/frai.2025.1569804.

[49] U. Maqsood, S. U. Rehman, T. Ali, K. Mahmood, T. Alsaedi, and M. Kundi, "An intelligent framework based on deep learning for sms and e-mail spam detection," *Applied Computational Intelligence and Soft Computing*, vol. 2023, no. 1, Jan. 2023, art. no. 6648970, doi: 10.1155/2023/6648970.

[50] A. Hadi, M. Mosleh, and K. Mohebbi, "Improved detection of email and sms spam using a hybrid deep and shallow learning model," *Scientific Reports*, vol. 16, no. 1, Oct. 2026, art. no. 33674, doi: 10.1038/s41598-026-74463-x.

[51] C. Opara, P. Modesti, and L. Golightly, "Evaluating spam filters and stylometric detection of ai-generated phishing emails," *Expert Systems with Applications*, vol. 276, Jun. 2025, art. no. 127044, doi: 10.1016/j.eswa.2025.127044.

[52] R. Gutierrez, W. Villegas-Ch, and J. Govea, "Cross-model evaluation of phishing detectors against llm-generated emails," *Frontiers in Big Data*, vol. 9, Jul. 2026, art. no. 1883452, doi: 10.3389/fdata.2026.1883452.

[53] R. Ghawa and A. Alhogail, "Hybrid ai-based detection of llm-generated phishing emails," *Electronics*, vol. 15, no. 15, Aug. 2026, art. no. 3383, doi: 10.3390/electronics15153383.

[54] T. B. Pan, Q. Yang, A. J. Cole, R. Wilson, and D. L. Woodard, "Psychology of phishing emails: Quantifying persuasion principles and simulating detection with large language models," *Expert Systems with Applications*, vol. 313, Jun. 2026, art. no. 131507, doi: 10.1016/j.eswa.2026.131507.

[55] S. K. Birthriya, P. Ahlawat, and A. K. Jain, "Detection and prevention of spear phishing attacks: A comprehensive survey," *Computers & Security*, vol. 151, Apr. 2025, art. no. 104317, doi: 10.1016/j.cose.2025.104317.

[56] G. Schwalbe and B. Finzel, "A comprehensive taxonomy for explainable artificial intelligence: a systematic survey of surveys on methods and concepts," *Data Mining and Knowledge Discovery*, vol. 38, no. 5, pp. 3043–3101, Sep. 2024, doi: 10.1007/s10618-022-00867-8.

[57] F. Bodria, F. Giannotti, R. Guidotti, F. Naretto, D. Pedreschi, and S. Rinzivillo, "Benchmarking and survey of explanation methods for black box models," *Data Mining and Knowledge Discovery*, vol. 37, no. 5, pp. 1719–1778, Sep. 2023, doi: 10.1007/s10618-023-00933-9.

[58] M. Saarela and V. Podgorelec, "Recent applications of explainable ai (xai): A systematic literature review," *Applied Sciences*, vol. 14, no. 19, Oct. 2024, art. no. 8884, doi: 10.3390/app14198884.

[59] S. Reynaud and A. Roxin, "Review of explainable artificial intelligence for cybersecurity systems," *Discover Artificial Intelligence*, vol. 5, no. 1, May 2025, art. no. 78, doi: 10.1007/s44163-025-00318-5.

[60] M. T. Ribeiro, S. Singh, and C. Guestrin, ""Why should I trust you?": Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, San Francisco, CA, USA, 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.

[61] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.

[62] S. M. Lundberg *et al.*, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020, doi: 10.1038/s42256-019-0138-9.

[63] V. Vimbi, N. Shaffi, and M. Mahmud, "Interpreting artificial intelligence models: a systematic review on the application of lime and shap in alzheimer's disease detection," *Brain Informatics*, vol. 11, no. 1, Dec. 2024, art. no. 10, doi: 10.1186/s40708-024-00222-1.

[64] M. A. Uddin, M. N. Islam, L. Maglaras, H. Janicke, and I. H. Sarker, "Explainabledetector: Exploring transformer-based language modeling approach for sms spam detection with explainability analysis," *Digital Communications and Networks*, vol. 11, no. 5, pp. 1504–1518, Oct. 2025, doi: 10.1016/j.dcan.2025.07.008.

[65] M. A. H. Prince, G. M. S. Hossain, T. Mahmud, M. A. Rahman, M. S. Hossain, and K. Andersson, "An explainable hybrid transformer-based feature fusion approach for smishing detection in cybersecurity," *IEEE Access*, vol. 14, pp. 106 124–106 147, 2026, doi: 10.1109/ACCESS.2026.3712173.

[66] M. A. Uddin, M. Mahiuddin, and I. H. Sarker, "An explainable transformer-based model for phishing email detection: A large language model approach," *Computer Networks*, vol. 277, Mar. 2026, art. no. 112061, doi: 10.1016/j.comnet.2026.112061.

[67] Q. Shambour, M. Al-Zyoud, and O. Almomani, "Quantum-inspired hybrid metaheuristic feature selection with shap for optimized and explainable spam detection," *Symmetry*, vol. 17, no. 10, Oct. 2025, art. no. 1716, doi: 10.3390/sym17101716.

[68] S. Panthi, M. Patidar, K. Zalawadia, D. Vekariya, and V. Vekariya, "A performance analysis of shap and lime for interpreting boosting models in phishing url detection," *Scientific Reports*, vol. 16, no. 1, Aug. 2026, art. no. 33730, doi: 10.1038/s41598-026-66800-x.

[69] T. Koide, H. Nakano, and D. Chiba, "Chatspamdetector: Large language model-based phishing email detection," *IEEE Access*, vol. 14, pp. 93 498–93 514, 2026, doi: 10.1109/ACCESS.2026.3704566.

[70] A. kadhim Jasim, F. A. Al-Ibraheemi, and H. A. Alkaabi, "Explainable ai for sms spam filtering: A novel hybrid architecture combining fuzzy logic and bidirectional lstm networks," *Franklin Open*, vol. 14, Mar. 2026, art. no. 100466, doi: 10.1016/j.fraope.2025.100466.

[71] J. DeYoung *et al.*, "ERASER: A benchmark to evaluate rationalized NLP models," in *Proc. 58th Annu. Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4443–4458, doi: 10.18653/v1/2020.acl-main.408.

[72] S. Jain and B. C. Wallace, "Attention is not explanation," in *Proc. Conf. North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, Minneapolis, MN, USA, 2019, pp. 3543–3556, doi: 10.18653/v1/N19-1357.

[73] H. Zhao *et al.*, "Explainability for large language models: A survey," *ACM Transactions on Intelligent Systems and Technology*, vol. 15, no. 2, pp. 1–38, Apr. 2024, doi: 10.1145/3639372.

[74] S. Krishna, T. Han, A. Gu, S. Wu, S. Jabbari, and H. Lakkaraju, "The disagreement problem in explainable machine learning: A practitioner's perspective," *Transactions on Machine Learning Research*, 2024.

[75] O. Mitruţ, G. Moise, A. Moldoveanu, F. Moldoveanu, M. Leordeanu, and L. Petrescu, "Clarity in complexity: how aggregating explanations resolves the disagreement problem," *Artificial Intelligence Review*, vol. 57, no. 12, Oct. 2024, art. no. 338, doi: 10.1007/s10462-024-10952-7.

[76] H. Baniecki and P. Biecek, "Adversarial attacks and defenses in explainable artificial intelligence: A survey," *Information Fusion*, vol. 107, Jul. 2024, art. no. 102303, doi: 10.1016/j.inffus.2024.102303.

[77] J. Kim, H. Maathuis, and D. Sent, "Human-centered evaluation of explainable ai applications: a systematic review," *Frontiers in Artificial Intelligence*, vol. 7, Oct. 2024, art. no. 1456486, doi: 10.3389/frai.2024.1456486.

[78] G. Romeo and D. Conti, "Exploring automation bias in human–ai collaboration: a review and implications for explainable ai," *AI & SOCIETY*, vol. 41, no. 1, pp. 259–278, Jan. 2026, doi: 10.1007/s00146-025-02422-7.

[79] A. K. Pathak, M. Gupta, and G. Jain, "Unmasking the clever hans effect in ai models: shortcut learning, spurious correlations, and the path toward robust intelligence," *Frontiers in Artificial Intelligence*, vol. 8, Jan. 2026, art. no. 1692454, doi: 10.3389/frai.2025.1692454.

[80] V. Dogra, S. Verma, Kavita, M. Woźniak, J. Shafi, and M. F. Ijaz, "Shortcut learning explanations for deep natural language processing: A survey on dataset biases," *IEEE Access*, vol. 12, pp. 26 183–26 195, 2024, doi: 10.1109/ACCESS.2024.3360306.

[81] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988, doi: 10.1016/0306-4573(88)90021-0.

[82] A. Mohammed and R. Kora, "A comprehensive review on ensemble deep learning: Opportunities and challenges," *Journal of King Saud University - Computer and Information Sciences*, vol. 35, no. 2, pp. 757–774, Feb. 2023, doi: 10.1016/j.jksuci.2023.01.014.

[83] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[84] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, San Francisco, CA, USA, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[85] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[86] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, 2020, art. no. 6, doi: 10.1186/s12864-019-6413-7.

[87] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[88] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947, doi: 10.1007/BF02295996.

[89] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988, doi: 10.2307/2531595.

[90] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[91] S. Luccioni, Y. Jernite, and E. Strubell, "Power hungry processing: Watts driving the cost of ai deployment?" in *Proc. ACM Conf. Fairness, Accountability, and Transparency (FAccT)*, Jun. 2024, pp. 85–99, doi: 10.1145/3630106.3658542.